

MUHAMMAD ABUBAKAR

192-A, P-II, Link Road, Model Town, Lahore | 0326-6966686 | muhammad.abubakar3520@gmail.com

SUMMARY

Software Engineer and Management Trainee Officer (IT) at Transsion Holdings with a strong foundation in Information Technology, full-stack web development, and machine learning. Experienced in providing technical support, troubleshooting hardware, software, and network systems, while developing responsive web applications using React and the MERN stack. Skilled in Python, JavaScript, HTML, CSS, and applied machine learning, including SVM classification and deep learning with ResNet34, EfficientNetB0, and attention mechanisms. Passionate about solving real-world problems, continuously learning emerging technologies, and building scalable, efficient software solutions.

TECHNICAL SKILLS

Web Development: HTML, CSS, JavaScript, React.js, Node.js, Express.js, MongoDB (MERN Stack), RESTful API design, responsive UI/UX design

Machine Learning & Data Science: Python, Support Vector Machines (SVM), Deep Learning (ResNet34, EfficientNetB0, CBAM attention module), transfer learning, computer vision & image processing, data augmentation, model evaluation (accuracy, precision, recall, F1-score, confusion matrix)

Programming Languages: Python, JavaScript, C++, C, Assembly (EMU 8086)

Tools & Platforms: Visual Studio Code, Google Colab, Jupyter Notebook

Soft Skills: Problem Solving, Critical Thinking, Teamwork, Time Management, Communication

PROFESSIONAL EXPERIENCE

Management Trainee Officer (MTO) – Information Technology

July 2026 – Present

Transsion Holdings (Infinix, TECNO & itel)

- Support IT operations and provide technical assistance across departments.
- Troubleshoot hardware, software, and network-related issues to ensure smooth business operations.
- Install, configure, and maintain computer systems, operating systems, and business applications.
- Assist with system administration, user account management, and IT asset management.
- Collaborate with senior IT professionals to implement and maintain IT infrastructure.
- Monitor system performance, resolve technical issues, and document support activities.
- Follow IT security policies and contribute to continuous process improvements.
- Gain hands-on experience with enterprise IT environments, technical support, and infrastructure management.

PROJECTS

Intelligent Voting Platform with AI Integration

MERN Stack

- Engineered a secure full-stack digital voting platform using MongoDB, Express.js, React.js, and Node.js, integrating multilayer biometric security (CNIC validation, facial recognition, fingerprint scanning).
- Developed OTP-based multi-factor authentication (MFA) to strengthen account security and prevent unauthorized access, building expertise in full-stack development and cybersecurity protocols.

Hospital Management System

React, MongoDB

- Built a full-stack hospital management web app (React front end, Node.js/MongoDB back end) with patient records, doctor profiles, appointment scheduling, and real-time updates.
- Designed a responsive, user-friendly interface with RESTful APIs and secure handling of sensitive medical data, strengthening database design and system integration skills.

Plant Leaf Disease Classification Using Deep Learning

ResNet34, EfficientNetB0, CBAM

- Built a multi-class plant disease classifier on the Plant Village dataset, implementing and comparing ResNet34, EfficientNetB0, and EfficientNetB0 with a Convolutional Block Attention Module (CBAM).
- Applied image preprocessing, data augmentation, and model optimization; the attention-based EfficientNetB0 model achieved the highest classification accuracy, evaluated via precision, recall, and F1-score.

Brain Tumor Detection using Machine Learning

SVM Classifier

- Built an MRI-based brain tumor detection model using a Support Vector Machine (SVM) classifier, preprocessing images and extracting key features to distinguish tumor from non-tumor cases.
- Strengthened skills in image processing, feature extraction, model training, and evaluation of classification performance on medical imaging data.

AI-Based Student Engagement Detection System

EfficientNetB0, XGBoost, TensorFlow

- Developed an AI-powered student engagement detection system to classify six engagement states (Focused, Confused, Frustrated, Bored, Looking Away, and Drowsy) from classroom images through transfer learning and feature extraction.
- Applied image preprocessing, feature extraction, model training, and evaluation using confusion matrix, precision, recall, and F1-score, achieving 98.76% training accuracy and 96.46% testing accuracy, while enabling future integration with webcam-based real-time classroom monitoring.

Tic Tac Toe Game on Console

Assembly Language – EMU 8086

- Developed a two-player console game handling turns, input validation, and win/draw logic, building a foundation in lowlevel programming and computer organization.

CERTIFICATIONS

- Machine Learning Internship – CodeAlpha
- Web Development Internship – Hex Softwares
- React Development
- Python

EDUCATION

BS in Computer Science

2022 – 2026

University of Management and Technology (UMT), Lahore

Relevant coursework: Programming Fundamentals, Object-Oriented Programming, Data Structures & Algorithms, Databases, Operating Systems, Artificial Intelligence, Data Science

Intermediate (FSc. Pre-Engineering)

2020 – 2022

Board of Intermediate and Secondary Education, Lahore

2018 – 2020

Marks obtained: 916

Matriculation (Science)

The New School, Lahore

Marks obtained: 965

LEADERSHIP & ACTIVITIES

Orientation Leader

UMT Fall Batch 2024

- Led campus orientation programs to help new BS students settle into university life.
- Organized activities at Juggi Taleem to engage children and promote community participation.

LANGUAGES

Fluent in Urdu, Punjabi, and English.